Joint Distributions

Joint PMF (discrete):

$$p_{X,Y}(x,y) = P(X = x, Y = y)$$

$$\sum_x \sum_y p_{X,Y}(x,y) = 1$$

Joint PDF (continuous):

$$f_{X,Y}(x,y)$$

$$\int \int f_{X,Y}(x,y)\,dx\,dy = 1$$

Marginal Distributions
How to find marginals from joint:
Discrete:

$$p_X(x) = \sum_y p_{X,Y}(x,y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x,y)$$

Continuous:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

—

Conditional Distributions

Definition (discrete):

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$$

Definition (continuous):

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

How to find conditional density step-by-step:
1. Start with joint density $f_{X,Y}(x,y)$ 2. Find marginal $f_X(x)$ by integrating out y 3. Divide:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Similarly,

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

—

Independence

X and Y are independent if:

- Discrete: $p_{X,Y}(x,y) = p_X(x)p_Y(y)$
- Continuous: $f_{X,Y}(x,y) = f_X(x)f_Y(y)$
- CDF: $F_{X,Y}(x,y) = F_X(x)F_Y(y)$
- Probabilities: $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$

If Independent ($X \perp Y$):

- $f_{Y|X}(y|x) = f_Y(y)$ (Conditional = Marginal)
- $E[Y|X] = E[Y]$ and $Var(Y|X) = Var(Y)$
- $E[XY] = E[X]E[Y]$ and $Cov(X,Y) = 0$
- $g(X) \perp h(Y)$ for any functions g, h

Checklist for dependence: Not independent if $f_{Y|X}(y|x)$, $E[Y|X]$, or $Var(Y|X)$ depend on X. *Note:* $Cov(X,Y) = 0 \implies$ independence (unless jointly normal).

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Expectation

Definitions:

- Discrete: $E[X] = \sum_x x p_X(x)$
- Continuous: $E[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx$
- Function $g(X,Y)$: $E[g(X,Y)] = \sum_x \sum_y g(x,y)p(x,y)$ or $\iint g(x,y)f(x,y)\,dx\,dy$

Linearity: $E[aX + bY + c] = aE[X] + bE[Y] + c$
Conditional Expectation:

- $E[Y|X = x] = \sum_y y\, p_{Y|X}(y|x)$ or $\int y\, f_{Y|X}(y|x)\,dy$
- $E[g(X)|Y = y] = \sum_x g(x)p_{X|Y}(x|y)$ or $\int g(x)f_{X|Y}(x|y)\,dx$
- **Tower Property (Extended):** $E[E[g(X,Y)|X]] = E[g(X,Y)]$
- **LIE Expansion:** $E[XY] = E[XE[Y|X]]$

—

Expectation & Variance Identities

Law of Total Expectation: $E[Y] = E[E[Y|X]]$ **Law of Total Variance:** $Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$ **Linearity:** $E[aX + bY + c] = aE[X] + bE[Y] + c$ **Scaling:** $Var(aX + b) = a^2Var(X)$ **Joint Var:** $Var(aX \pm bY) = a^2Var(X) + b^2Var(Y) \pm 2abCov(X,Y)$ **Independence:** If $X \perp Y$, then $Var(aX \pm bY) = a^2Var(X) + b^2Var(Y)$.

Transformations of Random Variables

If $Y = g(X)$ and g is monotone:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

CDF and PDF Relationship

$$F_X(x) = P(X \le x)$$

If continuous:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$$

—

Sampling Distributions & SE & CLT

Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance σ^2.
Sample Mean $\bar{X}$: $E[\bar{X}] = \mu$ (Unbiased), $Var(\bar{X}) = \frac{\sigma^2}{n}$, $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$
Difference/Sum of Sample Means (n_1, n_2):

- $Var(\bar{X}_1 \pm \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \pm 2\,\mathrm{Cov}(\bar{X}_1, \bar{X}_2)$
- **Cov of Means:** $\mathrm{Cov}(\bar{X}_1, \bar{X}_2) = \frac{1}{n_1 n_2} \sum_i \sum_j \mathrm{Cov}(X_i, Y_j)$

Sample Proportion $\hat{p}$: $E[\hat{p}] = p$ (Unbiased), $SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$
Central Limit Theorem (CLT): For large n (usually $n \ge 30$):

- $\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \implies Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \approx N(0,1)$
- $\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right)$ (if $np \geq 15, n(1-p) \geq 15$)

Estimator Properties:

- $Bias(\hat{\theta}) = E[\hat{\theta}] - \theta$
- $SE(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$ (used for CIs)
- $SE(\hat{p}) = \sqrt{p_0(1-p_0)/n}$ (used for Hypothesis Tests)

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Empirical Estimators

Sample mean:

$$\bar{x} = \frac{1}{n}\sum x_i$$

Sample variance:

$$s^2 = \frac{1}{n-1}\sum (x_i - \bar{x})^2$$

Sample covariance:

$$s_{XY} = \frac{1}{n-1}\sum (x_i - \bar{x})(y_i - \bar{y})$$

Sample correlation:

$$r = \frac{s_{XY}}{s_X s_Y}$$

—

Bayes' Theorem (General Form)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Continuous:

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)}$$

—

Inference: Confidence Intervals

General Form: CI = Estimator $\pm$ (Critical Value) $\times$ SE **Margin of Error (MoE):** MoE = (Critical Value) $\times$ $SE \implies$ Width $= 2 \times MoE$
Specific Confidence Intervals:

- **Mean (σ known):** $\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
- **Mean (σ unknown):** $\bar{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$
- **Proportion:** $\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Sample Size n for (MoE) or Length (L):

- **Mean:** $n = \left(\frac{z_{\alpha/2}\sigma}{MoE}\right)^2$ or $n = \frac{4z_{\alpha/2}^2\sigma^2}{L^2}$
- **Prop (Prior $\hat{p}$):** $n = \frac{z_{\alpha/2}^2\hat{p}(1-\hat{p})}{MoE^2}$ or $n = \frac{4z_{\alpha/2}^2\hat{p}(1-\hat{p})}{L^2}$
- **Prop (Conservative):** $n = \frac{z_{\alpha/2}^2}{4MoE^2}$ or $n = \frac{z_{\alpha/2}^2}{L^2}$

*Always **round UP** for n. Note: $L = 2 \times MoE$.*
Interpretation:

- **Long-run:** $100(1-\alpha)\%$ of theoretical intervals contain the true parameter.
- Higher confidence $\Rightarrow$ wider interval. Larger $n \Rightarrow$ narrower interval.
- Reject $H_0 : \theta = \theta_0$ at level α if $\theta_0 \notin (1-\alpha)100\%$ CI.

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Inference: Hypothesis Testing

Framework: H_0 (null) vs H_a (alternative). **Decision Rules:**

- p-value $\leq \alpha \implies$ Reject H_0. (Small p, reject null).
- Right-tail: $P(Z > z)$. Left-tail: $P(Z < z)$. Two-sided: $2P(Z > |z|)$.
- Rejection: Reject if $|stat| > z_{\alpha/2}$ (two-sided) or $stat > z_\alpha$ (right-tail).

Specific Tests:

- **Z-mean (σ known):** $Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$
- **T-mean (σ unknown):** $T = \frac{\bar{X}-\mu_0}{s/\sqrt{n}}$, $df = n-1$
- **Z-prop:** $Z = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$ (valid if $np_0 \geq 15, n(1-p_0) \geq 15$)

Error Types & Power:

- **Type I (α):** Reject H_0 when true.
- **Type II (β):** Fail to reject H_0 when false.
- **Power ($1-\beta$):** $P(\text{Reject } H_0|H_0 \text{ is false})$.

Normal Distribution Moments & Properties

If $Z \sim N(0,1)$:

- **Odd moments:** $E[Z^{2k+1}] = 0$
- **First four moments:** $E[Z] = 0, E[Z^2] = 1, E[Z^3] = 0, E[Z^4] = 3$

Linear Combinations: If $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$ are **Independent**:

$$aX + bY \sim N(a\mu_x + b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2)$$

Jointly Normal Independence: If (X,Y) jointly normal then $\text{Cov}(X,Y) = 0 \iff X \perp Y$.

Paired Data

Let $D = X_1 - X_2$ be the differences for n pairs.

- $E[D] = \mu_1 - \mu_2 = \mu_D$
- $Var(D) = \sigma_1^2 + \sigma_2^2 - 2Cov(X_1, X_2)$
- $Var(D) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$ (Shortcut if ρ given)
- **Insight:** If $\rho > 0$, variance is reduced (better efficiency).
- $s_D^2 = \frac{\sum(D_i - \bar{D})^2}{n-1}$
- **Sample s_D Shortcut:** $s_D = \sqrt{s_X^2 + s_Y^2 - 2(r)(s_X)(s_Y)}$

Paired T-test: $T = \frac{\bar{D}-\mu_{D,0}}{s_D/\sqrt{n}}$, $df = n-1$

Proportion Nuances

Testing vs CI:

- **Test** uses p_0 in $SE = \sqrt{p_0(1-p_0)/n}$ (assumes H_0 is true).
- **CI** uses $\hat{p}$ in $SE = \sqrt{\hat{p}(1-\hat{p})/n}$.

Worst-case variance: $p(1-p)$ is maximized at $p = 0.5$. $\Rightarrow$ Conservative n uses 0.25 in the numerator.

Final Exam Strategy

Common Mistakes to Avoid:

- Using $\hat{p}$ instead of p_0 in hypothesis tests.
- Forgetting $df = n - 1$ in t-tests.
- Forgetting the factor of 2 in two-sided p-values.
- Forgetting to **round UP** for sample size n.
- If there is a group, an n, or average use SE (reliability of the average) not sd(indiv.)

R Probability Functions: {prefix}{suffix}

- **Prefixes:**
  - `d`: Density (PDF/PMF height) [1, 2]
  - `p`: Cumulative Probability ($P(X \leq q)$) [1, 2]
  - `q`: Quantile/Inverse CDF (finds x given p) [1, 2]
  - `r`: Random Generation [3]
- **Suffixes (Distributions):**
  - `norm(q, mean, sd)`: Normal Distribution [2, 4]
  - `t(q, df)`: Student's t Distribution [5, 6]
  - `binom(x, size, prob)`: Binomial Distribution [7]
  - `unif(q, min, max)`: Uniform Distribution [2, 7]
  - **Critical Value (Z):** `qnorm(1 - alpha/2)` for two-sided [6].
  - **Critical Value (T):** `qt(1 - alpha/2, df)` for two-sided.
  - *Example CL 96.5% (n=33):* `qt(p=0.9825, df=32)`
- **Descriptive & Utility:**
  - `mean()`, `sd()`, `var()`, `IQR()`, `quantile(x, p)` [3, 5, 8]
  - `summary()`, `table()`, `prop.table()`, `round(val, n)` [1-3, 9]
  - `rowSums()`, `colSums()`, `sum()` [10, 11]

R Functions: Simple Linear Regression

- *model* $\leftarrow$ *lm*($Y \sim X$): Fits a simple linear regression model of response `Y` on predictor `X` [1].
- `summary(model)`: Extracts the coefficient estimates ($\hat{\beta}_0, \hat{\beta}_1$), their standard errors, t-values, p-values, the residual standard error ($\hat{\sigma}$), and the Multiple R^2 [1].
- `anova(model)`: Generates the ANOVA table, including Regression Sum of Squares (SSReg), Error Sum of Squares (SSError), Mean Squares (MS), and the Global F-statistic [1].

Additional Formulas & Properties

Two Independent Means ($\mu_1 - \mu_2$)

Equal Variances: $S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$, $df = n_1 + n_2 - 2$
$SE = S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$, $T = \frac{(\bar{X}_1-\bar{X}_2)-\Delta_0}{SE}$, CI: $(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2;df}SE$
Check Rule: Assume equal if $\frac{\max(S_1^2,S_2^2)}{\min(S_1^2,S_2^2)} < 5$ ($n \approx 7$), < 3 ($n \approx 15$), or < 2 ($n \approx 30$).

Unequal Variances: $SE = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$, $T = \frac{(\bar{X}_1-\bar{X}_2)-\Delta_0}{SE}$, CI: $(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2;\nu}SE$
$df = \nu = \frac{(S_1^2/n_1+S_2^2/n_2)^2}{(S_1^2/n_1)^2/(n_1-1)+(S_2^2/n_2)^2/(n_2-1)}$

Two Independent Proportions ($p_1 - p_2$)

Test ($H_0 : p_1 = p_2$): Pooled $\hat{p} = \frac{n_1\hat{p}_1+n_2\hat{p}_2}{n_1+n_2}$, $SE = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$
CI & Test ($H_0 : p_1 - p_2 = \Delta_0 \neq 0$): $SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
$Z = \frac{(\hat{p}_1-\hat{p}_2)-\Delta_0}{SE}$, CI: $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}SE$

Simple Linear Regression ($Y = \beta_0 + \beta_1 X + \epsilon$)

Estimates: $\hat{\beta}_1 = \frac{S_{X,Y}}{S_X^2} = r_{X,Y}\frac{S_Y}{S_X}$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}$
Residuals ($\hat{\epsilon}_i$): $\sum \hat{\epsilon}_i = 0$, $\sum X_i\hat{\epsilon}_i = 0$, $\sum \hat{Y}_i\hat{\epsilon}_i = 0$. Line passes through $(\bar{X}, \bar{Y})$.
ANOVA: $SSTotal = SSReg + SSError$, $df : n - 1 = 1 + (n - 2)$
$R^2 = \frac{SSReg}{SSTotal} = 1 - \frac{SSError}{SSTotal}$, $\hat{\sigma}^2 = MSError = \frac{SSError}{n-2}$
Std. Errors: $SE(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{(n-1)S_X^2}}$, $SE(\hat{\mu}_Y(x)) = \hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x-\bar{X})^2}{\sum(X_i-\bar{X})^2}}$
F-test ($H_0 : \beta_1 = 0$): $F = \frac{MSReg}{MSError} = T^2$

Concept Quick Hits

Power: $1 - P(\text{Type II Error})$. Probability of correctly rejecting a false H_0.
CI Width Trade-offs: $n \uparrow \Rightarrow$ narrower width. Confidence Level $(1 - \alpha) \uparrow \Rightarrow$ wider width.
CI Probability: After sampling, the probability the CI contains truth is exactly 0 or 1.